GREEN INNOVATION PITCHING COMPETITION

ANNEX B: Concept Paper Template

Team Name	Pew Pew
Higher Education Institution	University of Science and Technology of Southern Philippines (USTP) - CDO
Challenge Area	[X] A. Waste to Worth [ ] B. Green Schools and Learning [ ] C. Resilient and Sustainable Communities
Stage of Development	[ ] Idea / concept [X] Prototype [ ] Early pilot

1. Title of the Innovation

SubAI: AI-Assisted Drainage Triage and Verified Plastic Recovery Tracker for Cagayan de Oro

Tagline: Sa SubAI, magtinabangay tang kanunay aron ang basura atong masubay. (In subAI, let us always help one another so we can track our waste. "Subay" means to track.)

2. The Problem

Anyone living in Cagayan de Oro knows what happens during an ordinary afternoon storm. Within 30 to 45 minutes of heavy rain exceeding 35 mm per hour, vital thoroughfares turn into dangerous floodways. Water rises knee to waist deep across Carmen along J.V. Serina and Max Suniel Streets, Lapasan along Claro M. Recto Avenue, Kauswagan along the national highway, and commercial alleys around Cogon Market. Ground floor classrooms and stores are inundated, public transit along Sayre Highway stalls completely, and thousands of commuters and students are stranded.

The root cause is that surface curb inlets, catch basins, and culvert grates are choked with floating solid waste. Official 2024 to 2025 baseline figures provided directly to our team by the Research, Planning and Development Division of the City Local Environment and Natural Resources Office (CLENRO) show that Cagayan de Oro generates 157,000 metric tons of solid waste annually (430.14 tons per day). Recyclables make up 21 percent of this total (32,970.07 tons per year or 90.33 tons per day), while residual packaging represents another 23 percent (36,110.07 tons per year). City waste flow studies indicate that 11 percent of all plastics escape regular collection as unmanaged street litter. In flood prone communities like Carmen (generating 16,642 tons annually, including 3,495 tons of recyclables) and Lapasan (generating 8,573 tons, including 1,800 tons of recyclables near the 11-kilometer Bitan-ag Creek outlet), unmanaged plastic bottles and food packaging choke drainage grates and reduce hydraulic intake capacity by up to 75 percent.

Compounding this crisis, municipal maintenance remains entirely reactive and linear. City crews like CLENRO's Hapsay Sapa team are dispatched only after streets are submerged and traffic is paralyzed. When workers clear out clogged culverts, the retrieved debris is treated as worthless muck and hauled unsegregated to the overflowing Pagalungan Sanitary Landfill, which has faced regulatory notices from DENR-EMB Region 10. This cycle wastes tons of clean recyclable polymers and leaves communities exposed to the next storm. (Sources: RPD-CLENRO 2024 to 2025 Waste Inventory; CDRRMD Inundation Reports; DPWH Region 10 Drainage Master Plan; DENR-EMB Region 10 Landfill Audits).

3. The Solution

subAI is an installable, low-bandwidth Progressive Web App (PWA) that converts everyday drainage complaints into a verified, closed-loop waste recovery workflow in five direct steps:

1. Citizen Capture and Auto-Geolocate (<15 Seconds): When a resident, commuter, or student spots an obstructed curb inlet, they snap a photo. The client browser compresses the image in under 100 milliseconds to under 200 kilobytes for instant upload over mobile networks, while one-tap GPS automatically locks coordinates and tags the local barangay.
2. Multimodal AI Triage (<1.5 Seconds): Google Gemini Multimodal Vision classifies the structure (Curb Inlet, Catch Basin, Culvert, or Open Canal) and auto-detects visible debris categories (PET bottles, flexible packaging/sachets, silt, organics). Maintenance coordinators review the alert, eliminate duplicate submissions, and generate an objective 0 to 5 priority score based on flood hazard.
3. Digital Work Order Dispatch: The system dispatches an actionable digital task directly to the mobile devices of CLENRO Hapsay Sapa crews or barangay maintenance squads, complete with navigation and pre-identified material tags.
4. Curb Sorting and Digital Scale Weighing: Field personnel clear the grate, upload an after-cleanup verification photo, and weigh segregated materials on heavy-duty digital scales, recording exact kilograms for PET plastics, flexible films, silt, and residual waste.
5. Verified Recycler Handover and Mass-Balance Ledger: Segregated recyclables transfer directly to accredited Barangay Materials Recovery Facilities (MRFs) or junk-shop cooperatives, which confirm custody via digital on-screen weight receipts. The platform maintains a strict mass-balance ledger (Total Collected = Recycled + Landfill Residual + Pending Handover), ensuring zero unweighed estimates ever count as recycled mass.

Unlike reactive cleanup alone, subAI links citizen alert, response time, location, before and after proof, and scale-verified disposition in a single transparent audit trail.

4. Circular Economy Model

subAI operationalizes the internationally recognized circular economy framework while directly strengthening the municipal Materials Recovery Facility (MRF) network mandated under Republic Act No. 9003 through five core strategies:

1. Rethink: Replaces the conventional view of stormwater drains as waste dump chutes, transforming them into decentralized passive interception points that trap valuable polymers before they degrade into waterways.
2. Recover: Rather than allowing waste to mix into contaminated sludge, cleanup crews perform point-of-extraction separation at the curb, isolating PET bottles, caps, flexible films, organics, and silt.
3. Recycle and MRF Integration: Scale-verified clean PET plastics are delivered directly to accredited Barangay MRFs (such as Carmen, Lapasan, and Bulua) and registered junk-shop cooperatives, where they are mechanically shredded and pelletized into raw manufacturing resin.
4. Repurpose: Low-value multi-layer sachets and flexible plastic films that commercial junk shops reject are channeled to partner upcycling enterprises to be compressed into durable eco-bricks and paving blocks.
5. Reduce: Civic hotspot intelligence maps provide City Hall and barangay councils with empirical evidence to enact targeted single-use plastic reduction policies and community education.

Together, these five strategies keep materials in use longer and prevent plastic leakage into Bitan-ag Creek and Macajalar

Bay while ensuring honest, scale-audited material recovery. (Sources: Republic Act No. 9003; Republic Act No. 11898; UN SDGs 12 and 14).

5. Technology and AI Component

Technology Architecture: The prototype uses Next.js 16 App Router, React 19, Tailwind CSS v4, Firebase Cloud Firestore with secure server actions, Leaflet with OpenStreetMap mapping, and an installable Progressive Web App (PWA). This architecture eliminates app store friction, enabling citizens and municipal workers to launch the system instantly on budget mobile devices.

Artificial Intelligence Engine: We integrate Google Gemini Multimodal Vision API via secure server actions. When an obstruction photo is uploaded, the vision model analyzes visual features to classify drainage structures (Curb Inlet, Catch Basin, Culvert, Open Canal) and suggest debris fractions in 1.5 seconds, saving maintenance staff manual data entry time.

Data Needs and Privacy: The AI requires only a single downscaled photo (under 960 by 960 pixels, under 200 kilobytes) focused on the drainage inlet. It collects no facial or biometric data. In compliance with the Philippine Data Privacy Act of 2012 (RA 10173), user location coordinates are generalized on public dashboards to protect personal privacy.

Managing Bias and Hallucination: The AI serves strictly as an advisory suggestion tool; human reviewers confirm or override every tag with a single tap. If an image is blurry or network service drops, the system defaults smoothly to simple manual dropdown menus. Crucially, the platform never allows AI to estimate kilograms from photographs; actual material recovery numbers are proven exclusively by physical digital scales.

6. Target Users and Beneficiaries

Target Users:
1. Citizens, Students, and Commuters: The city's 815,779 residents and daily visitors act as eyes on the ground, snapping geotagged photo reports in under 15 seconds.
2. Maintenance and Sanitation Teams: About 120 field personnel across CLENRO's Hapsay Sapa crews, CDRRMD Oro Rescue units, barangay tanods, and school facilities staff receive prioritized digital work orders on their phones and log weighed cleanup evidence.
3. Local Recyclers and MRF Operators: Barangay materials recovery facilities and registered junk-shop cooperatives receive sorted plastics and confirm official digital handover receipts.
Direct Beneficiaries: Over 250,000 residents, commuters, and small business owners living along chronically flooded areas in Carmen, Lapasan, Kauswagan, Cogon, and Bulua gain safer, flood-free streets and protected shops. Academic institutions like USTP, Capitol University, Liceo de Cagayan, and Xavier University, as well as critical hospitals like Northern Mindanao Medical Center (NMMC), benefit from unhampered emergency access. Ultimately, the entire coastal population around Macajalar Bay benefits as thousands of kilograms of plastic are trapped on land before polluting marine waters.

7. Implementation and Sustainability Plan

Estimated Year 1 Pilot Budget (PHP 85,000):
• Digital Cloud Systems (PHP 15,000): Serverless hosting on Vercel, Firebase Cloud Firestore subscriptions, custom domain, and Gemini API operational buffer.
• Verification Hardware (PHP 25,000): Three heavy-duty digital platform and hanging scales, weatherproof mobile field clipboards, and protective gear for cleanup crews.
• Community and Barangay Onboarding (PHP 30,000): Practical training workshops for CLENRO staff and barangay tanods, plus durable aluminum QR code plates installed near recurring drainage hotspots so residents can scan and report instantly.
• Maintenance and Contingency (PHP 15,000): Scale calibration, physical tool repairs, and field logistics.
Strategic Partners: CLENRO for citywide maintenance alignment, CDRRMD for flood hazard coordination, the City Education and Development Office (CEDO), the Association of Barangay Captains (ABC) for community adoption, USTP for academic research backing, and local recycling cooperatives.
Long-Term Financial Sustainability: (1) Municipal and Campus Licensing: Providing subAI as an affordable maintenance management platform for neighboring local government units like El Salvador and Tagoloan, as well as university campuses. (2) Aggregated Recyclable Sales: Consolidating verified, clean PET plastic volumes from multiple barangays to sell directly to industrial recyclers at premium bulk rates. (3) Extended Producer Responsibility (EPR) Auditing: Generating third-party verified, scale-backed plastic diversion data for consumer goods companies seeking compliance under Republic Act No. 11898.

8. Expected Impact and Measures of Success

During our twelve-month pilot across high-risk drainage corridors in Cagayan de Oro, subAI targets three tangible, realistic measures of success:
1. 5,000 Kilograms of Waste Diverted: Intercepting at least 5,000 kilograms of drainage solid waste away from landfills and waterways, including at least 2,000 kilograms of PET bottles and 1,200 kilograms of flexible plastic films, certified by physical digital scale weight slips.
2. 60 Percent Faster Clearing Response: Slashing average drain clearance response times from 48 hours down to under 6 hours during storm alerts, effectively eliminating chronic standing water at 12 critical urban street bottlenecks.
3. Active Civic Adoption and Full Auditability: Engaging over 2,500 registered community reporters, resolving more than 800 verified work orders, and maintaining a 100 percent reconciled mass balance where every single kilogram of collected waste is accounted for.

9. SDGs Addressed

Core to Waste to Worth: SDG 11 (Sustainable Cities and Communities): Protects low-income and vulnerable neighborhoods from flash floods, clears hazardous standing water, and modernizes municipal solid waste collection; SDG 12 (Responsible Consumption and Production): Substantially reduces waste generation through proactive drain interception, source segregation, and verified recycling handovers; SDG 14 (Life Below Water): Prevents plastic debris from traveling down Bitan-ag and Iponan creeks into Macajalar Bay, protecting marine biodiversity and coastal fishing communities.
Supporting Goals: SDG 9 (Industry, Innovation and Infrastructure): Delivers an accessible, low-bandwidth digital public infrastructure for municipal sanitation teams; SDG 13 (Climate Action): Strengthens local adaptive capacity against severe monsoon downpours by linking proactive drainage clearing with rooftop rainwater storage; SDG 17 (Partnerships for the Goals): Fosters strong collaboration among local government agencies, college students, barangay tanods, and private recycling enterprises.

10. Team Members and Roles

Name	Degree Program and Year	Role in the Team
Danlord Farell A. Soriano	1st Year BS Computer Science	Team Leader
Justine Roy P. Salvador	1st Year BS Computer Science	Lead Developer
John Kurt D. Montero	1st Year BS Computer Science	UI/UX Designer
Shawn Michael E. Hitalada	1st Year BS Computer Science	Concept Paper + Research
Mark Anthony S. Bacus	1st Year BS Computer Science	Presentation + Pitch

Faculty Coach: Isaiah Marc R. Postrero (Department of Computer Science, USTP CDO)

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4. Department of Public Works and Highways (DPWH Region X). (2024). Urban Drainage Master Plan for Cagayan de Oro City.
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